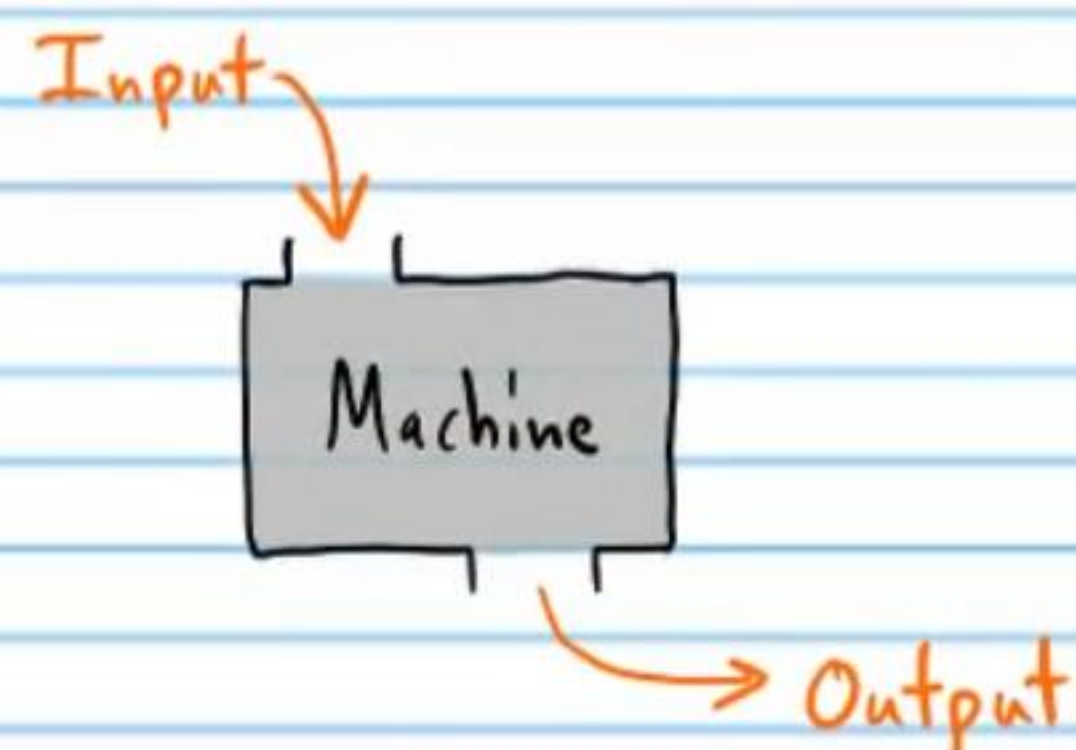


Review on Functions and Function Notation

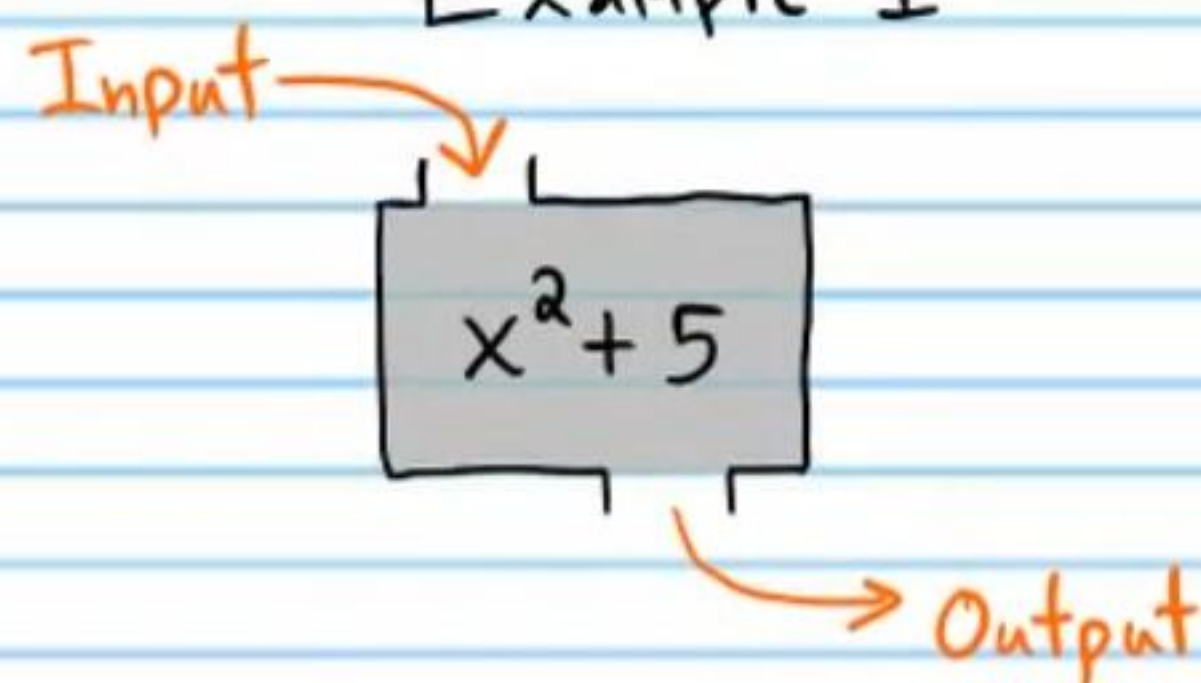
The General Concept of a Function

A function can be thought of as a machine.

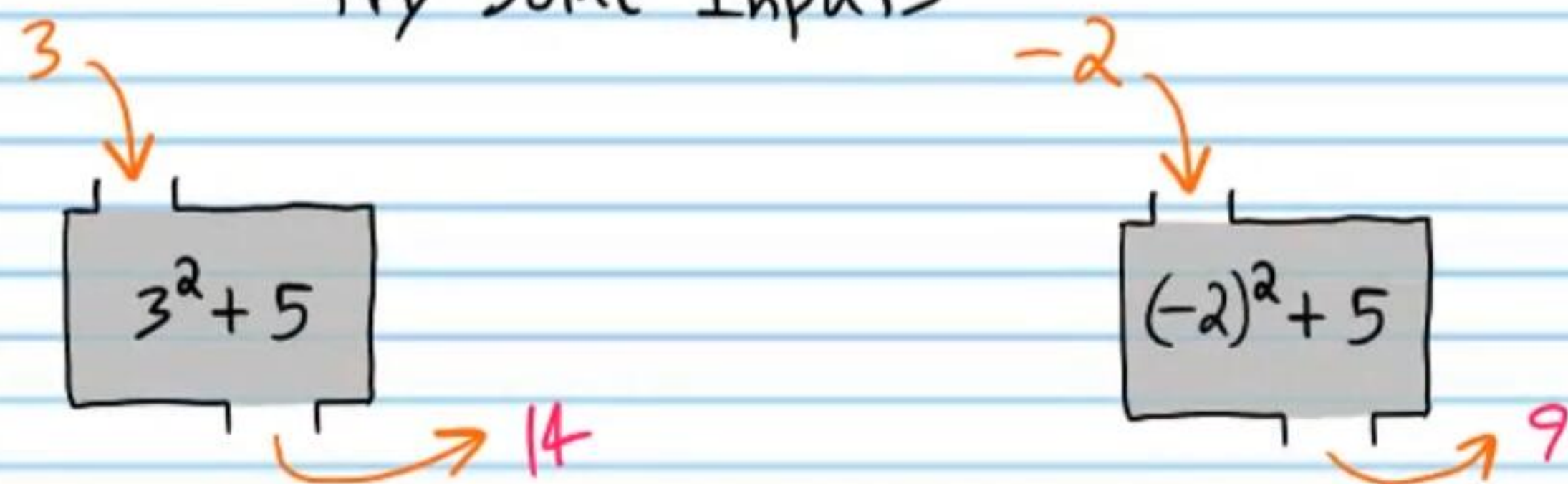


In the subject of mathematics, a function is a set of operations that are performed on a number.

Example I

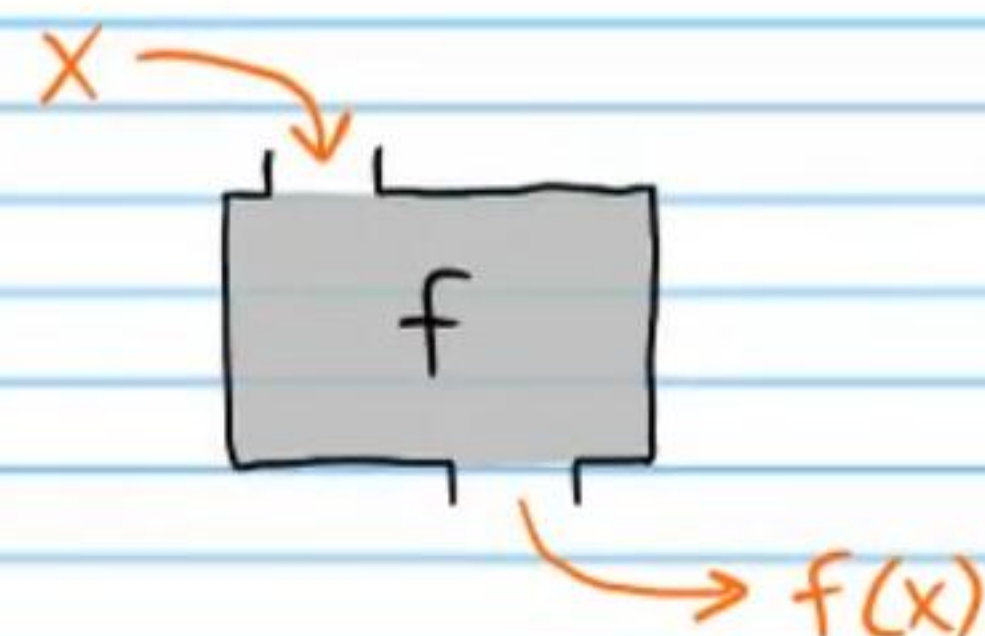


Try Some Inputs



Function Notation

"f" is the most common letter used for function notation, "x" is the most common letter used for inputs, and $f(x)$ represents an output resulting from an input "x".

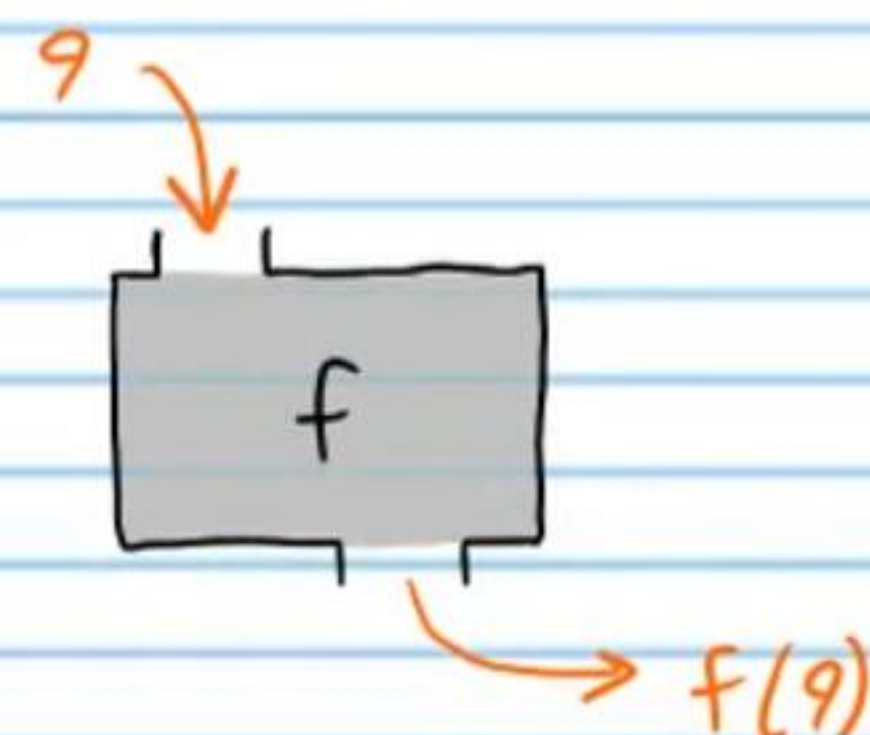


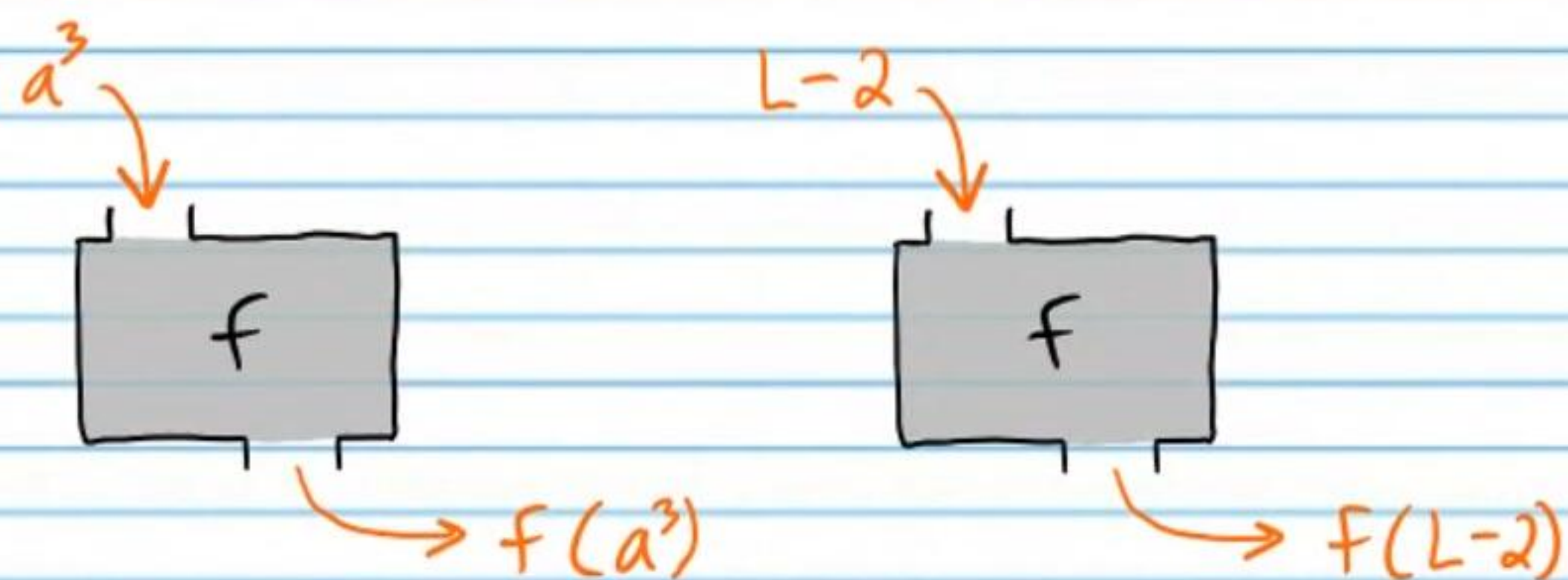
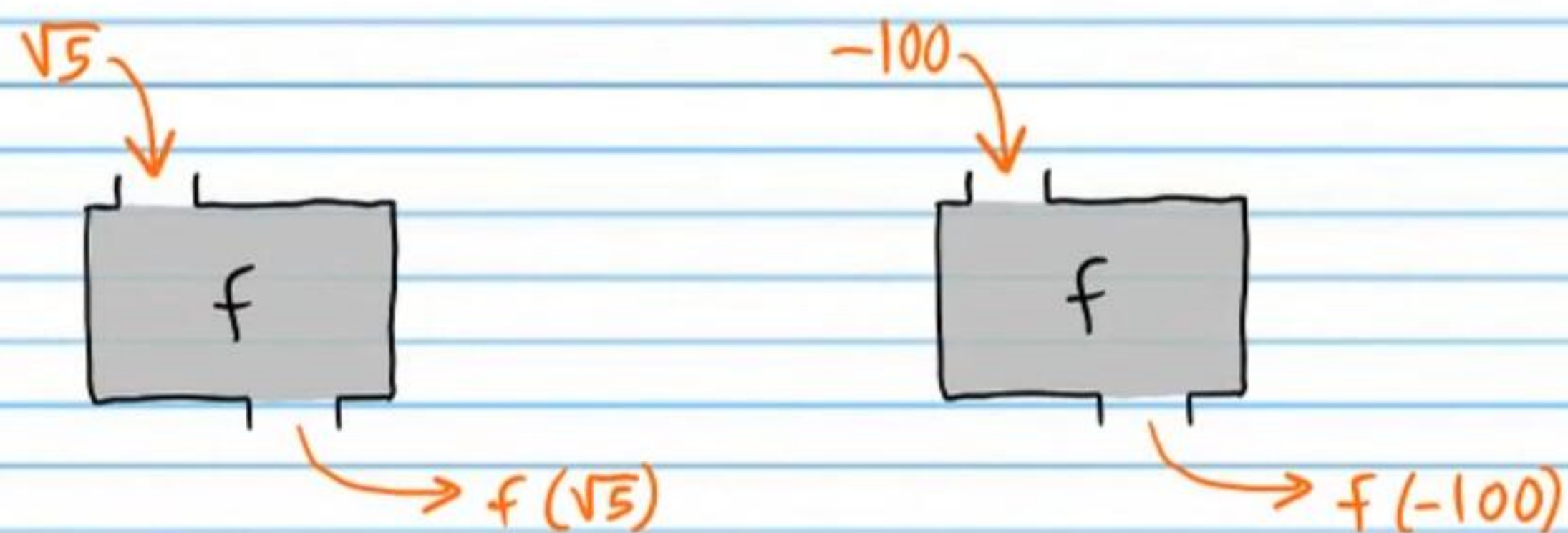
$f(x)$ is pronounced "f of x."

Warning: Although this notation is most commonly used, it can be very confusing because the parentheses on $f(x)$ DO NOT indicate multiplication.

Using $f(x)$ Notation

Example II





Practice

- ① If $f(x) = 2x - 5$, find i) $f(3)$, ii) $f(-4)$, and iii) $f(0)$.

i) $f(3) = 2(3) - 5 = 6 - 5 = \boxed{1}$

ii) $f(-4) = 2(-4) - 5 = -8 - 5 = \boxed{-13}$

iii) $f(0) = 2(0) - 5 = \boxed{-5}$

- ② If $f(x) = \frac{10}{x} + 2$, fill in the table below.

x	f(x)
-5	0
-2	-3
4	4.5
0	Und

i) $f(-5) = \frac{10}{-5} + 2 = -2 + 2 = \boxed{0}$

ii) $f(-2) = \frac{10}{-2} + 2 = -5 + 2 = \boxed{-3}$

iii) $f(4) = \frac{10}{4} + 2 = \frac{5}{2} + 2 = \boxed{4.5}$

iv) $f(0) = \frac{10}{0} + 2 = \boxed{\text{Undefined}}$

- ③ If $f(x) = 2^x - 9$, find i) $f(2)$, ii) $f(3)$, and iii) $f(-1)$.

i) $f(2) = 2^2 - 9 = 4 - 9 = \boxed{-5}$

ii) $f(3) = 2^3 - 9 = 8 - 9 = \boxed{-1}$

iii) $f(-1) = 2^{-1} - 9 = \frac{1}{2} - 9 = \boxed{-8\frac{1}{2}}$

- ④ If $f(x) = 3 - 4|x|$, fill in the table below.

x	-3	0	π	-0.6
f(x)	-9	3	$3 - 4\pi$	0.6

i) $f(-3) = 3 - 4|-3| = 3 - 4(3) = 3 - 12 = \boxed{-9}$

ii) $f(0) = 3 - 4|0| = 3 - 4(0) = 3 - 0 = \boxed{3}$

iii) $f(\pi) = 3 - 4|\pi| = \boxed{3 - 4\pi}$

iv) $f(-0.6) = 3 - 4|-0.6| = 3 - 4(0.6) = \boxed{0.6}$

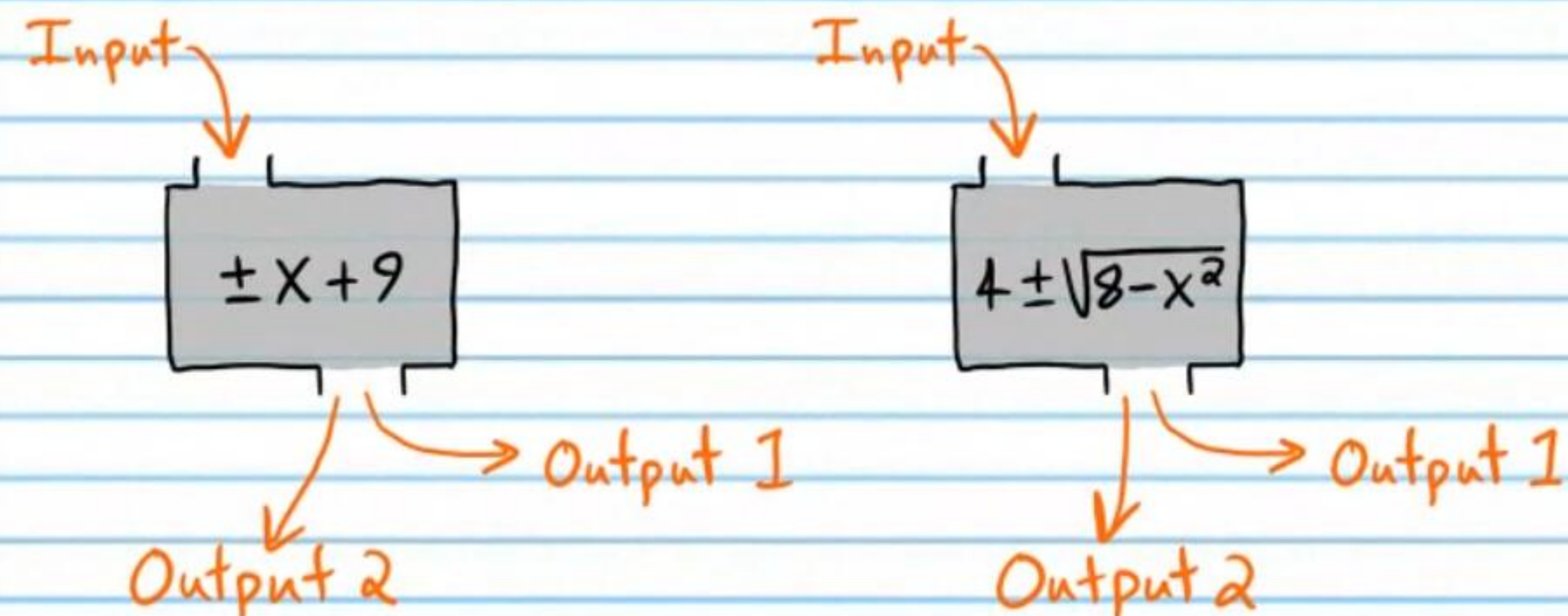
A Necessary Characteristic of Functions

In order for an expression to qualify as a function, every input must produce **ONLY ONE** output.

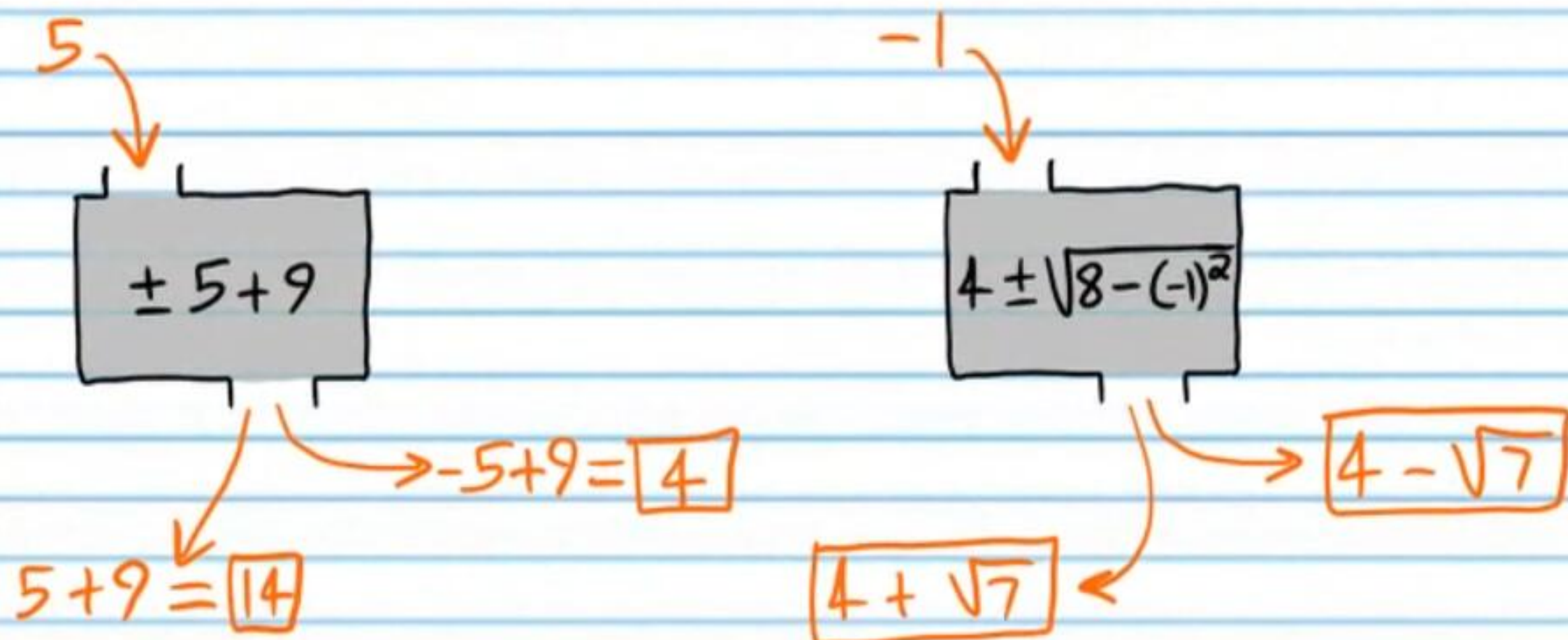
These are not functions because the expressions have ($\pm$) symbols on them, which indicate that for some inputs, there are two resulting outputs.

Example III

Example IV



Let's try some inputs.



In the tables below, x represents the inputs and y represents the outputs. Circle the tables that could not represent functions.

⑤

x	y
3	-2
8	10
0	4
-8	-7
1	5
-2	0

x	y
0	1
3	5
-7	-8
4	3
0	9
1	6

x	y
36	40
7	38
20	26
85	20
-100	10
67	9

x	y
7	16
7	5
8	2
-2	4
9	11
10	3

⑥

x	y
-4	2
8	-3
10	12
13	17
10	1/3
-4	1

x	y
0	13
2	3
-3	17
6	8
-3	10
6	-5

x	y
-7	2
8	4
10	5
-9	3
-2	-1
0	0

x	y
200	500
300	600
400	700
500	800
600	900
700	1000

⑦

x	y
1	2
5	8
2	13
4	0
9	1
-1	7

x	y
-7	8
9	2
-6	-3
-7	10
2	0
3	5

x	y
14	-17
8	-15
-9	0
-2	3
20	-16
31	-11

x	y
2	-3
4	-2
9	0
9	1
2.3	8
-0.7	0.9

8)

x	y
-1	-30
0	-20
1	-10
8	-3
9	5
12	-9

x	y
1	10
2	12
3	14
4	16
5	18
6	20

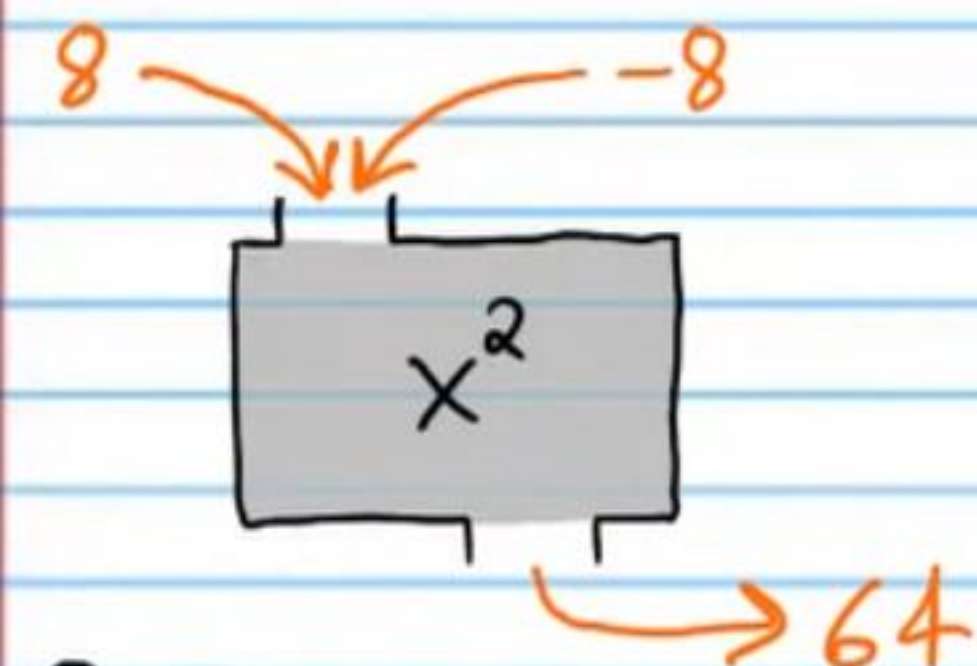
x	y
42	45
56	23
68	43
20	40
56	21
20	39

x	y
3	10
8	9
3	8
-3	7
-8	6
8	5

Warning: If different inputs produce the same output in an expression (Example V), then this does not disqualify the expression from being a function.

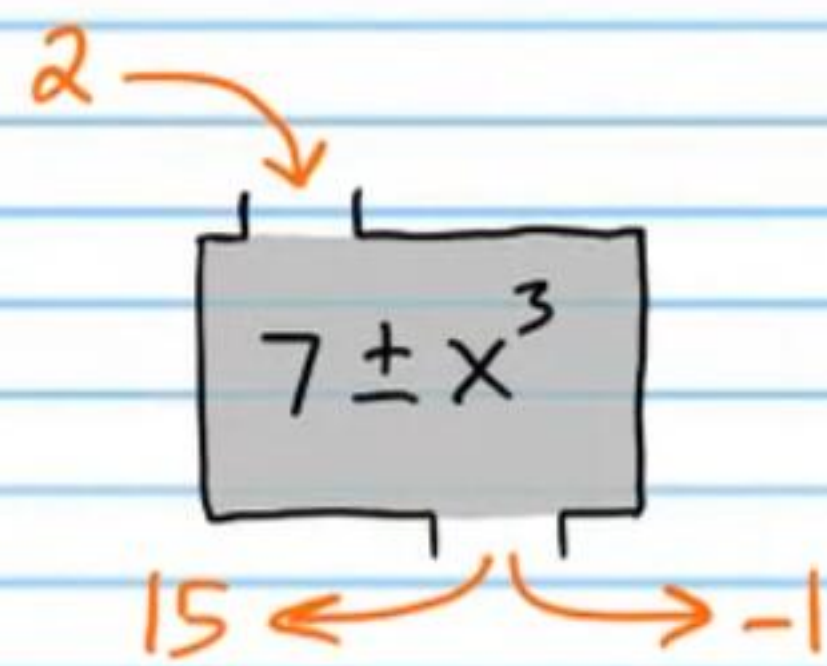
Example V

★ Can be a function.



Example VI

★ Cannot be a function.



9)

x	6	0	-2	3	10	19
y	3	8	9	-2	5	1

x	-1	2	4	-5	6	-5
y	3	8	8	1	0	-1

x	3	5	6	0	-8	10
y	-4	-1	2	2	3	5

x	-3	-8	9	10	12	13
y	1	1	1	2	3	1

10)

x	y
-7	-5
2	-5
3	-5
4	-5
0	4
-7	-5

x	y
4	3
8	3
9	0
10	-1
11	-2
11	-2

x	y
8	0
8	0
8	0
4	7
3	6
-2	5

x	y
5	2
6	3
5	2
6	8
4	-2
2	0

11)

x	10	-7	10	9	10	3
y	8	4	3	1	7	6

x	6	2	-1	17	11	12
y	4	3	3	0	0	4

x	y
-1	2
-8	0
-5	2
3	2
9	1
10	7

x	y
7	8
6	8
5	8
4	8
3	8
2	8

12)

x	20	3	7	-1	0	7
y	0	8	-4	2	18	-4

x	10	5	4	2	4	10
y	-10	8	2	1	3	-10

x	y
1	1
2	2
3	3
4	4
5	5
6	6

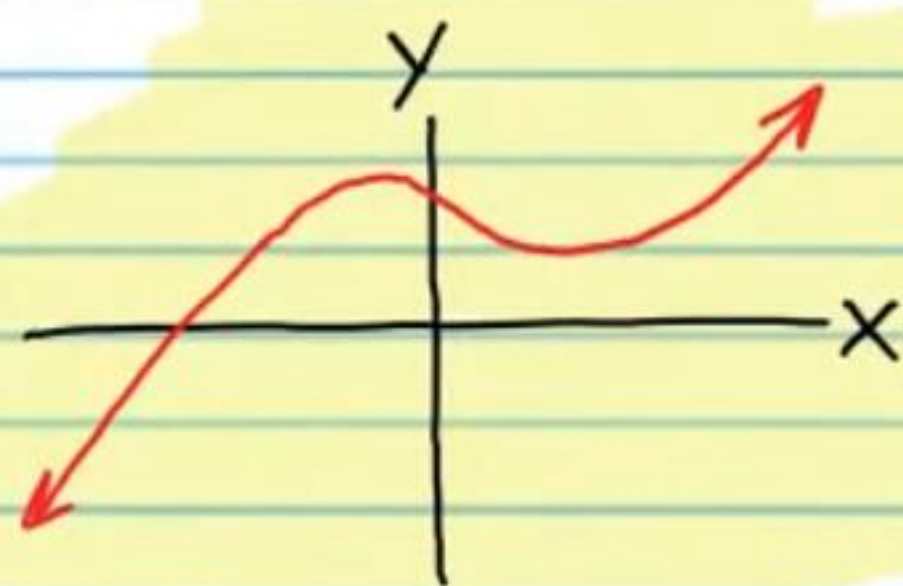
x	y
-3	9
-3	9
-2	0
-1	-1
0	0
1	2

Three Ways to Represent a Function

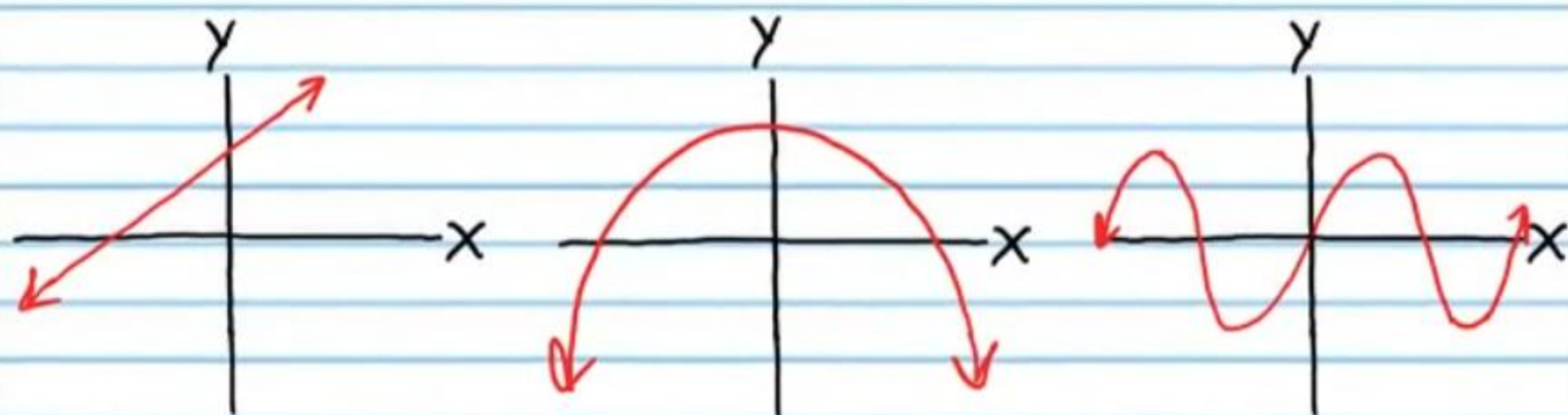
I. Equation

II. Table

III. Graph

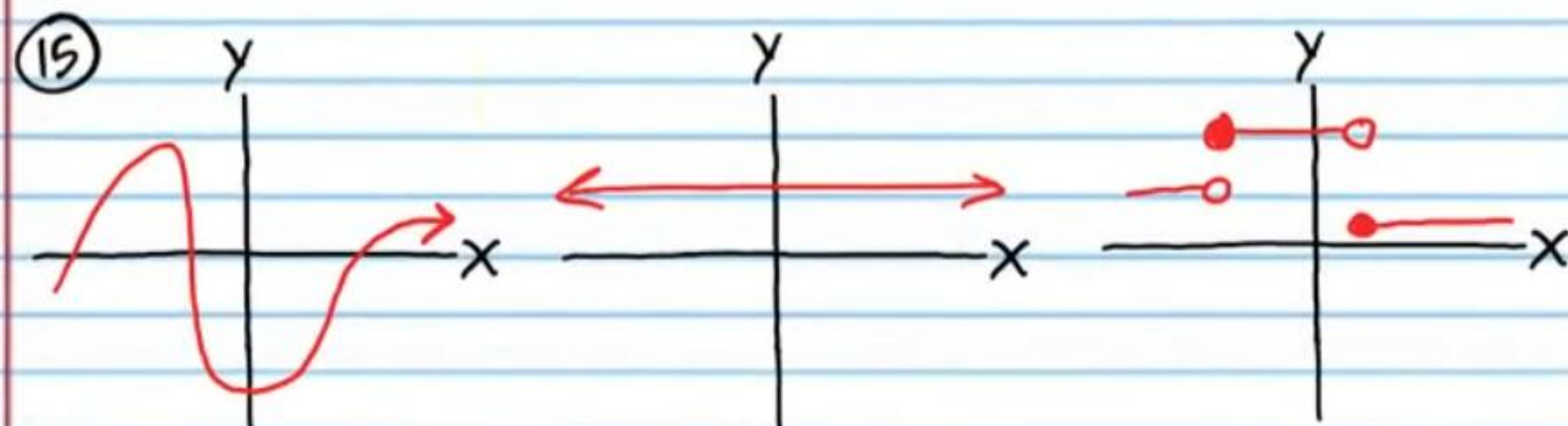
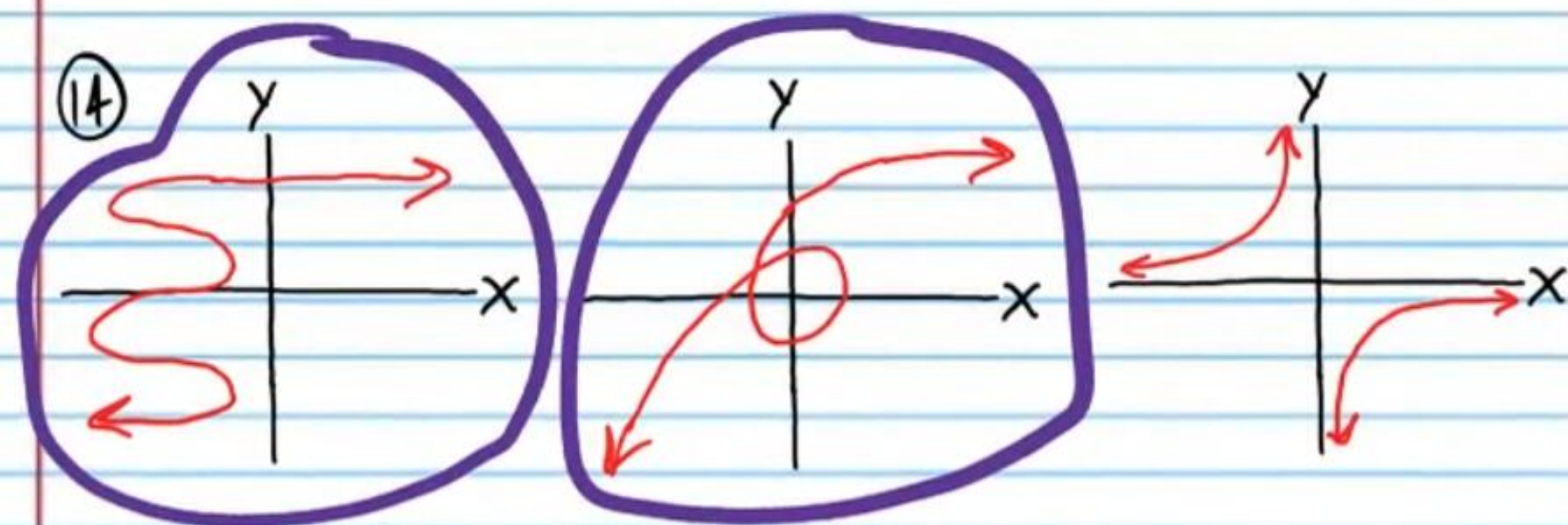
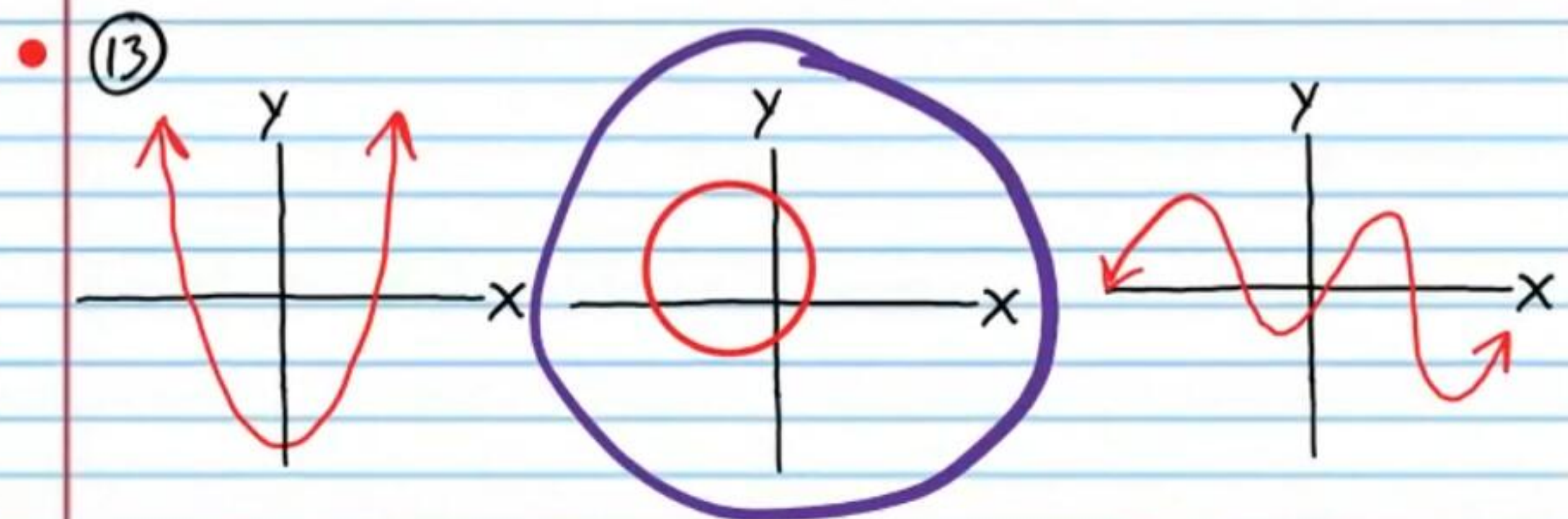


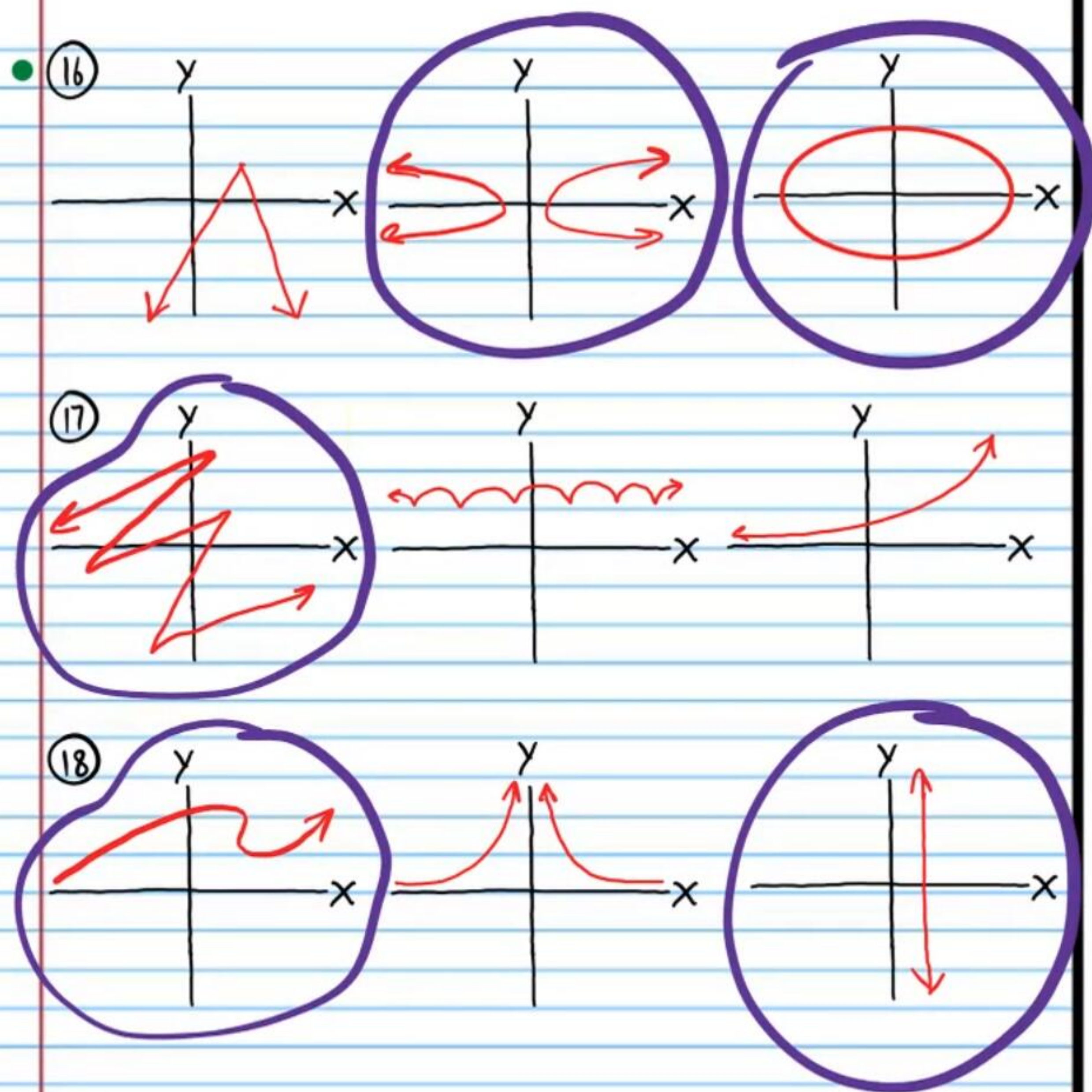
Examples of Graphs of Functions



The Vertical Line Test: If any vertical line touches a graph more than once, then the graph cannot be a function.

Circle the relations below that cannot be functions.





Function Notation Can Take Many Forms

- ⑯ If $\Delta x = x^2 + 90$, find $\Delta 3$, $\Delta -2$, $\Delta 7$, and $\Delta -1$.

$$\Delta 3 = 3^2 + 90 = 9 + 90 = \boxed{99}$$

$$\Delta -2 = (-2)^2 + 90 = 4 + 90 = \boxed{94}$$

$$\Delta 7 = 7^2 + 90 = 49 + 90 = \boxed{139}$$

$$\Delta -1 = (-1)^2 + 90 = 1 + 90 = \boxed{91}$$

- ⑰ If $A \odot B = A^3 + 2^B$, find $-1 \odot 5$, $2 \odot 6$, $-4 \odot 1$, and $3 \odot 3$.

$$-1 \odot 5 = (-1)^3 + 2^5 = -1 + 32 = \boxed{31}$$

$$2 \odot 6 = 2^3 + 2^6 = 8 + 64 = \boxed{72}$$

$$-4 \odot 1 = (-4)^3 + 2^1 = -64 + 2 = \boxed{-62}$$

$$3 \odot 3 = 3^3 + 2^3 = 27 + 8 = \boxed{35}$$

- ⑱ If $\Delta y = \frac{4}{y}$, find $\Delta 2$, $\Delta 3$, and $\Delta -1$.

$$\Delta 2 = \frac{4}{-2} = \boxed{-2}$$

$$\Delta 3 = \boxed{\frac{4}{3}}$$

$$\Delta -1 = \frac{4}{-1} = \boxed{-4}$$

②② If $\begin{matrix} B \\ \text{shaded box} \\ W \end{matrix} = 2B - 5W$, find $\begin{matrix} -10 \\ \text{shaded box} \\ 7 \end{matrix}$ and $\begin{matrix} 3 \\ \text{shaded box} \\ 4 \end{matrix}$.

$$\begin{matrix} -10 \\ \text{shaded box} \\ 7 \end{matrix} = 2(-10) - 5(7) = -20 - 35 = \boxed{-55}$$

$$\begin{matrix} 3 \\ \text{shaded box} \\ 4 \end{matrix} = 2(3) - 5(4) = 6 - 20 = \boxed{-14}$$

Three Letters That Are Most Commonly Used for Function Notation

Letters	f	g	h
Examples	$f(x) = 2x - 1$	$g(x) = 5^{x+4}$	$h(x) = 3(x-6)^2$

• ②③ If $f(x) = \frac{1}{8-2x}$, find $f(-6)$.

$$f(-6) = \frac{1}{8-2(-6)} = \frac{1}{8+12} = \boxed{\frac{1}{20}}$$

②④ If $g(x) = 4^{x-1}$, find $g(-2)$.

$$g(-2) = 4^{-2-1} = 4^{-3} = \frac{1}{4^3} = \boxed{\frac{1}{64}}$$

②⑤ If $h(x) = \frac{x^2}{3}$, find $h(1.2)$.

$$h(1.2) = \frac{1.2^2}{3} = \frac{1.44}{3} = \boxed{0.48}$$

• ②⑥ If $f(x) = x + 7$, find $f(3)$.

$$f(3) = 3 + 7 = \boxed{10}$$

②⑦ If $g(x) = \frac{3}{2}x + 10$, find $g(-4)$.

$$g(-4) = \frac{3}{2}(-4) + 10 = -6 + 10 = \boxed{4}$$

②⑧ If $h(x) = \sqrt{5x}$, find $h(\frac{1}{5})$.

$$h(\frac{1}{5}) = \sqrt{5(\frac{1}{5})} = \sqrt{1} = \boxed{1}$$

Piecewise Functions

Piecewise Function: A function defined by multiple expressions, each with its own **exclusive** interval of inputs.

Example: $f(x) = \begin{cases} x-2, & 1 < x < 5 \\ 4+x^2, & 5 \leq x < 10 \end{cases}$

Given the functions below, find the values.

• (29) $f(x) = \begin{cases} 2^x, & 0 < x < 4 \\ 5x-1, & 4 \leq x < 9 \end{cases}$

i) $f(3) = 2^3 = \boxed{8}$

ii) $f(6) = 5(6)-1 = 30-1 = \boxed{29}$

iii) $f(1) = 2^1 = \boxed{2}$

iv) $f(5) = 5(5)-1 = 25-1 = \boxed{24}$

v) $f(8) = 5(8)-1 = 40-1 = \boxed{39}$

• (30) $f(x) = \begin{cases} \frac{1}{x+2}, & -10 \leq x < -6 \\ x^3, & -4 \leq x \leq -2.1 \end{cases}$

i) $f(-9) = \frac{1}{-9+2} = \boxed{-\frac{1}{7}}$

ii) $f(-6) = \boxed{\text{Undefined}}$

iii) $f(-5) = \boxed{\text{Undefined}}$

iv) $f(-2.1) = (-2.1)^3 = \boxed{-9.261}$

• (31) $f(x) = \begin{cases} 3+x, & 2 < x < 7 \\ 4x, & 7 \leq x < 100 \end{cases}$

i) $f(9) = 4(9) = \boxed{36}$

ii) $f(5) = 3+5 = \boxed{8}$

iii) $f(30) = 4(30) = \boxed{120}$

iv) $f(6.5) = 3+6.5 = \boxed{9.5}$

• (32) $f(x) = \begin{cases} \frac{1}{2}x, & -4 < x < 8 \\ \sqrt{x+7}, & 10 \leq x \leq 15 \end{cases}$

i) $f(-4) = \frac{1}{2}(-4) = \boxed{-2}$

ii) $f(8) = \boxed{\text{Undefined}}$

iii) $f(-6) = \boxed{\text{Undefined}}$

iv) $f(0) = \frac{1}{2}(0) = \boxed{0}$

• (33) $f(x) = \begin{cases} -7(4^x)+1, & x < 2 \\ 4(2^x)-8, & x \geq 2 \end{cases}$

i) $f(3) = 4(2^3)-8 = 4(8)-8 = 32-8 = \boxed{24}$

ii) $f(20) = \boxed{4(2^{20})-8}$

iii) $f(-6) = -7(4^{-6})+1 = -7\left(\frac{1}{4^6}\right)+1 = \frac{-7}{4,096}+1 = \frac{4,089}{4,096}$

• (34) $f(x) = \begin{cases} 2, & x < -3 \\ 5+2x, & x > 5 \end{cases}$

i) $f(-2) = \boxed{\text{Undefined}}$

ii) $f(7) = 5+2(7) = 5+14 = \boxed{19}$

iii) $f(-6) = \boxed{2}$

iv) $f(-6.01) = \boxed{2}$

• ③⑤ $f(x) = \begin{cases} 1 + 3^{x-2}, & x \leq -1 \\ x + 4, & x > -1 \end{cases}$

i) $f(-2) = 1 + 3^{-2-2} = 1 + 3^{-4} = 1 + \frac{1}{3^4} = 1 + \frac{1}{81} = \boxed{1\frac{1}{81}}$

ii) $f(-1) = 1 + 3^{-1-2} = 1 + 3^{-3} = 1 + \frac{1}{3^3} = 1 + \frac{1}{27} = \boxed{1\frac{1}{27}}$

iii) $f(0) = 0 + 4 = \boxed{4}$

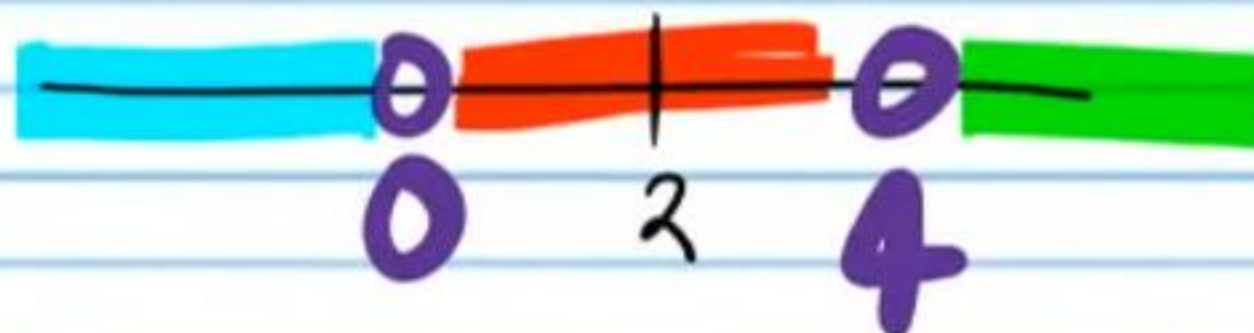
iv) $f(12) = 12 + 4 = \boxed{16}$

③⑥ $f(x) = \begin{cases} x^2 + 5x + 1, & x < 0 \\ 2x - 3, & x > 4 \end{cases}$

i) $f(7) = 2(7) - 3 = 14 - 3 = \boxed{11}$

ii) $f(2) = \boxed{\text{Undefined}}$

iii) $f(-100) = (-100)^2 + 5(-100) + 1 = 10,000 - 500 + 1 = \boxed{9,501}$



Notation Indicating Operations between Functions

If "f" and "g" are functions, then

$$(f + g)(x) = f(x) + g(x)$$

$$(f - g)(x) = f(x) - g(x)$$

$$(f \cdot g)(x) = f(x) \cdot g(x)$$

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$$

$$(f \circ g)(x) = f(g(x))$$

• ③⑦ If $f(x) = x^2 + 5x$ and $g(x) = 3x - 2$, find

A. $(f + g)(x) = f(x) + g(x) = x^2 + 5x + 3x - 2 = \boxed{x^2 + 8x - 2}$

B. $(f - g)(x) = f(x) - g(x) = x^2 + 5x - (3x - 2) = x^2 + 5x - 3x + 2 = \boxed{x^2 + 2x + 2}$

C. $(f \cdot g)(x) = f(x)g(x) = \boxed{(x^2 + 5x)(3x - 2)}$

$$D. \left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \frac{x^2+5x}{3x-2}$$

- ③⑧ If $f(x) = x+1$ and $g(x) = 3^x - 6$, find

$$A. (f+g)(x) = f(x) + g(x) = x+1 + 3^x - 6 = 3^x + x - 5$$

$$B. (f-g)(x) = f(x) - g(x) = x+1 - (3^x - 6) = x+1 - 3^x + 6 = -3^x + x + 7$$

$$C. (f \cdot g)(x) = f(x)g(x) = (x+1)(3^x - 6)$$

$$D. \left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \frac{x+1}{3^x - 6}$$

- ③⑨ If $f(x) = x+1$ and $g(x) = 6x$, find

$$A. (f \circ g)(x) = f(g(x)) = f(6x) = 6x+1$$

$$B. (g \circ f)(x) = g(f(x)) = g(x+1) = 6(x+1) = 6x+6$$

- ④⑩ If $f(x) = (x-2)^3 + x$ and $g(x) = 5x$, find

$$A. (f \circ g)(x) = f(g(x)) = f(5x) = (5x-2)^3 + 5x$$

$$B. (g \circ f)(x) = g(f(x)) = g((x-2)^3 + x) = 5[(x-2)^3 + x] = 5(x-2)^3 + 5x$$

- ④⑪ If $f(x) = \frac{9}{6-x}$ and $g(x) = -2x$, find

$$A. (f \circ g)(x) = f(g(x)) = f(-2x) = \frac{9}{6-(-2x)} = \frac{9}{6+2x}$$

$$B. (g \circ f)(x) = g(f(x)) = g\left(\frac{9}{6-x}\right) = -2\left(\frac{9}{6-x}\right) = -\frac{18}{6-x} = \frac{18}{x-6}$$

- ④⑫ If $f(x) = x^2 + 4x - 1$ and $g(x) = 2^x + 1$, find

$$A. (f \circ g)(x) = f(g(x)) = f(2^x + 1) = (2^x + 1)^2 + 4(2^x + 1) - 1$$

$$B. (g \circ f)(x) = g(f(x)) = g(x^2 + 4x - 1) = 2^{x^2 + 4x - 1} + 1$$